

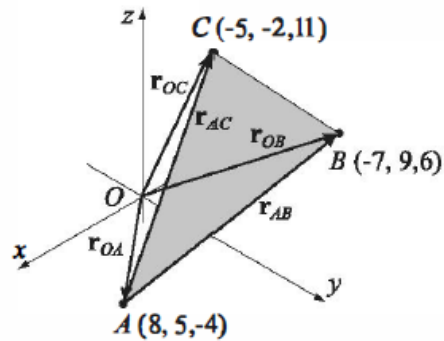
Chapter 3 Homework: Mathematical Operations with Arrays

Instructions:

- Write a script file for the exercises below.
- Make sure to write comments in your code and separate output display for clarity.
- Submit your script file in the following format: Ch3HW.m

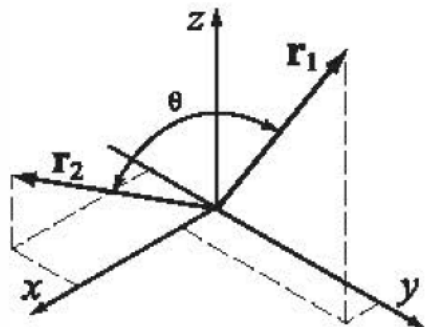
1.

The area of a triangle ABC can be calculated by $|\mathbf{r}_{AB} \times \mathbf{r}_{AC}|/2$, where $\mathbf{r}_{AB}$ and $\mathbf{r}_{AC}$ are vectors connecting the vertices A and B and A and C , respectively. Determine the area of the triangle shown in the figure. Use the following steps in a script file to calculate the area. First, define the vectors $\mathbf{r}_{OA}$, $\mathbf{r}_{OB}$ and $\mathbf{r}_{OC}$ from knowing the coordinates of points A , B , and C . Then determine the vectors $\mathbf{r}_{AB}$ and $\mathbf{r}_{AC}$ from $\mathbf{r}_{OA}$, $\mathbf{r}_{OB}$ and $\mathbf{r}_{OC}$. Finally, determine the area by using MATLAB's built-in functions `cross`, `sum`, and `sqrt`.



2. The dot product can be used for determining the angle between two vectors:

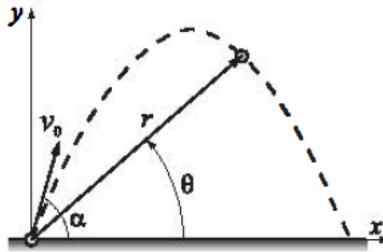
$$\theta = \cos^{-1} \left(\frac{\mathbf{r}_1 \cdot \mathbf{r}_2}{|\mathbf{r}_1| |\mathbf{r}_2|} \right)$$



Use MATLAB's built-in functions `acosd`, `sqrt`, and `dot` to find the angle (in degrees) between $\mathbf{r}_1 = 6\mathbf{i} - 3\mathbf{j} + 2\mathbf{k}$ and $\mathbf{r}_2 = 2\mathbf{i} + 9\mathbf{j} + 10\mathbf{k}$. Recall that $|\mathbf{r}| = \sqrt{\mathbf{r} \cdot \mathbf{r}}$.

3. The position function as a function of time $(x(t), y(t))$ of a projectile fired with a speed of v_o at an angle α is given by

$$x(t) = v_o \cos(\alpha) \cdot t \quad y(t) = v_o t \sin \alpha - \frac{1}{2} g t^2$$



where $g = 9.81 \text{ m/s}^2$. The polar coordinates of the projectile at time t are $(r(t), \theta(t))$, where $r(t) = \sqrt{x^2(t) + y^2(t)}$ and $\tan \theta(t) = \frac{y(t)}{x(t)}$. Consider the case where $v_o = 162 \text{ m/s}$ and $\alpha = 70^\circ$. Determine $r(t)$ and $\theta(t)$ for $t = 1, 6, 11, \dots, 31$ second.

4. Solve the following system of linear equations:

$$2.5a - b + 3c + 1.5d - 2e = 57.1$$

$$3a + 4b - 2c + 2.5d - e = 27.6$$

$$-4a + 3b + c - 6d + 2e = -81.2$$

$$2a + 3b + c - 2.5d + 4e = -22.2$$

$$a + 2b + 5c - 3d + 4e = -12.2$$

Output:

Problem 1:

```
Area =  
112.4433
```

Problem 2:

```
theta =  
86.9897
```

Problem 3:

```
r =  
1.0e+03 *  
0.1574    0.8083    1.2410    1.4759    1.5564    1.5773    1.7176  
theta =  
69.3893   65.7152   60.5858   53.0831   41.6187   24.0270   0.1812
```

Problem 4:

```
result =  
8.2000  
-2.0000  
4.8000  
6.0000  
-5.6000
```